IGCSE · Cambridge (CIE) · Biology

30 mins 9 questions

Target Test

Created 29/05/26

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Total Marks /30

1 (a) Animal and plant cells have some structural features in common.

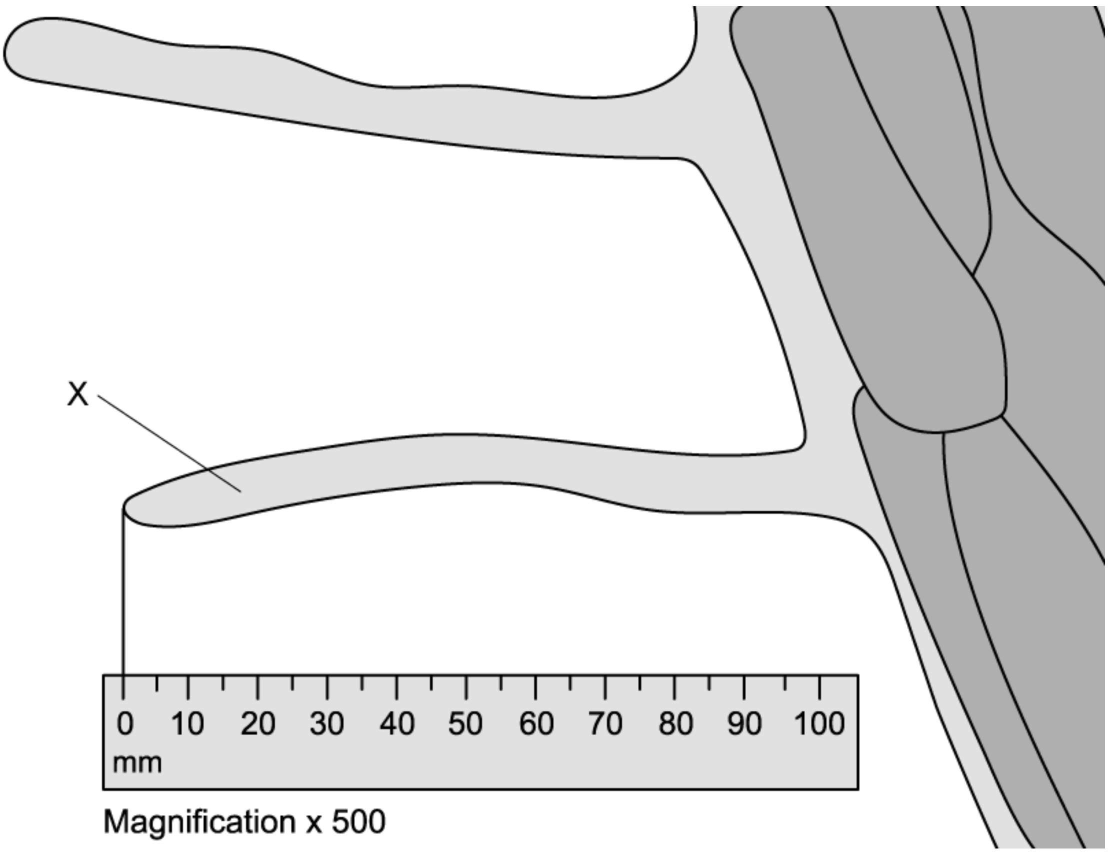
Name **two** structures you would find in both an animal cell and a plant cell.

....

.... **(2 marks)**

(b) **Fig.1** shows a photomicrograph of the surface of a plant root.

Fig. 1



Name structure **X**.

.... **(1 mark)**

(c) Calculate the actual length of structure **X** in **Fig. 1.**

Show your working.

....

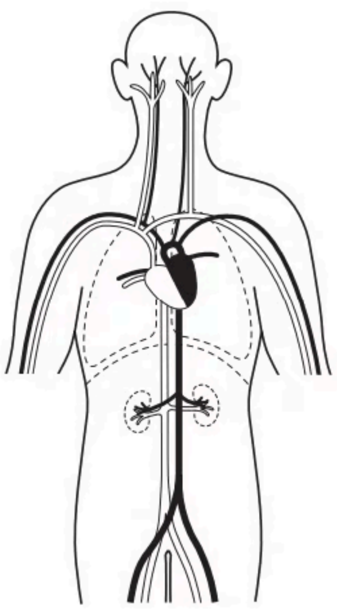
(2 marks)

(d) Failure of structure **X** to develop properly in the cells of a plant root could be catastrophic to a plant.

Suggest an explanation as to why.

(4 marks)

2 The diagram shows some of the blood vessels and other structures in the human body.



The blood vessels shown are all parts of the same.

A. cell.

B. organ.

C. organ system.

D. tissue.

(1 mark)

3 (a) The boxes on the left contain the names of structures in the body.

The boxes on the right contain the names of processes carried out by the body.

Draw one straight line from each structure to the process in which it is involved.

Draw **six** lines.

structure	process
aorta	breathing
cervix	circulation
duodenum	digestion
ribs	excretion
sensory neurone	reflex action
ureter	reproduction

(6 marks)

(b) State the functions of these **two** specialised cells.

Red blood cell ....

Ciliated cell ....

(2 marks)

(c) Fig. 1 shows a specialised plant cell.

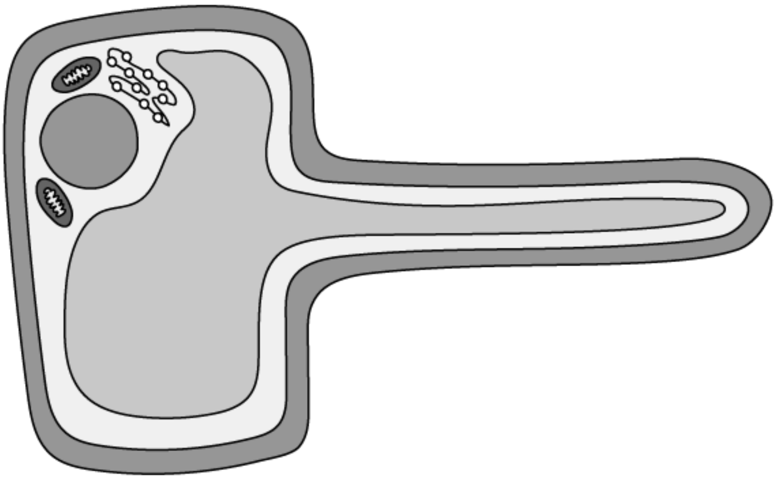


Fig. 1

(i)

Identify the plant cell in Fig. 1.

[1]

(ii)

State and explain **one** adaptation that allows this cell to carry out its function.

[2]

(3 marks)

(d) Identify with a tick (✓) the structures that are found in both plant and animal cells.

Structure	In animal cells?	In plant cells?
Nucleus		
Permanent vacuole		
Cell wall		

(3 marks)

4 An image of a cell is 24 mm long. The actual cell length is 6 μm.

What is the magnification?

A. x4

B. x40

C. x400

D. x4000

(1 mark)

5 A palisade mesophyll cell is found near the upper surface of a leaf and contains many chloroplasts.

Why is this an effective adaptation?

A. It reduces water loss by closing stomata.

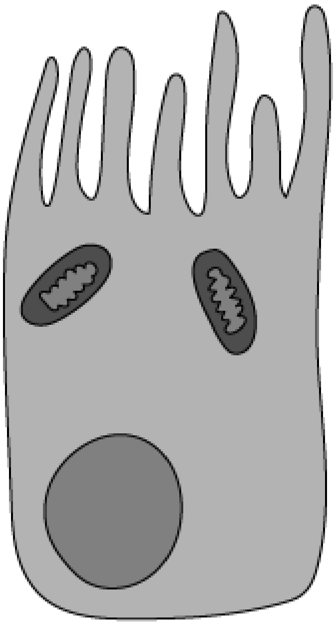
B. It allows the cell to absorb more light for photosynthesis.

C. It increases the surface area for gas exchange.

D. It allows the cell to respire at a higher rate.

(1 mark)

6 The diagram below shows part of an organism (ciliated epithelium).



What term would correctly describe the level of organisation of the structure in the diagram above?

A. Cell

B. Organelle

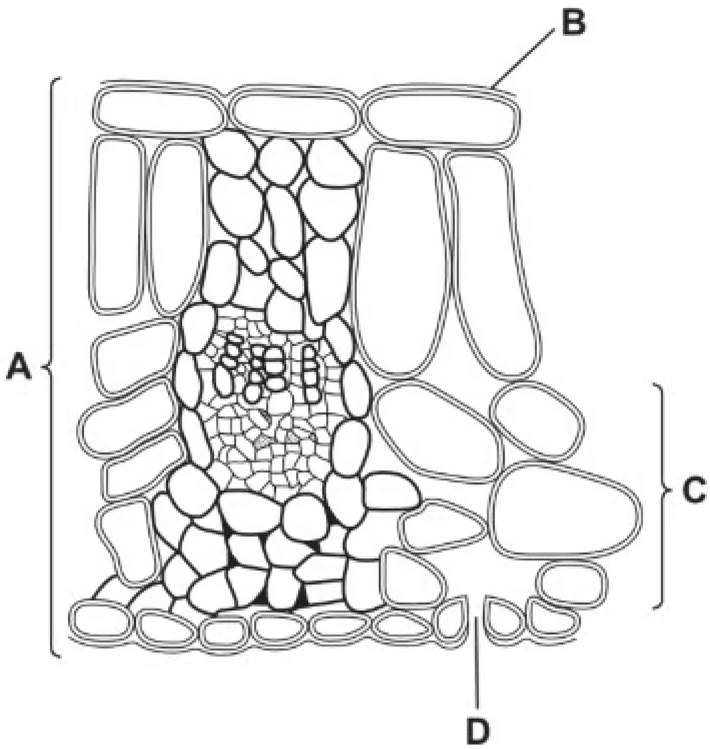
C. Organ

D. Organ system

(1 mark)

7 The diagram shows a cross-section through a leaf.

Which label shows a tissue?



(1 mark)

8 A photograph shows a plant cell nucleus measuring 2 mm across.

If the magnification of the cell is x500, what is the actual size of the nucleus?

A. 0.00002 mm

B. 0.004 mm

C. 0.04 mm

D. 250 mm

(1 mark)

9 Which row of the table below correctly matches a structure to its level of organisation within an organism?

	tissue	organ	system
A	phloem	small intestine	digestive
B	red blood cell	xylem	stomach
C	small intestine	sperm	digestive
D	root hair	leaf	heart

(1 mark)